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Frequency mixing circuit with multi-transistor architecture

Abstract

A frequency mixing circuit includes: a first transistor and a second transistor. The first transistor has a control terminal, a first terminal and a second terminal. The control terminal of the first transistor is configured to receive an oscillation signal, the first terminal of the first transistor is configured to output a mixed signal, and the second terminal of the first transistor is configured to receive a source signal. The second transistor has a control terminal, a first terminal and a second terminal. The control terminal of the second transistor is coupled to the second terminal of the first transistor, the first terminal of the second transistor is coupled to the first terminal of the first transistor, and the second terminal of the second transistor is coupled to the control terminal of the first transistor.

Inventors:	Wu; Yi-Ching (HsinChu, TW), Chang; Chia-Jun (HsinChu, TW)
Applicant:	Realtek Semiconductor Corp. (HsinChu, TW)
Family ID:	90833202
Assignee:	Realtek Semiconductor Corp. (HsinChu, TW)
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6130574	12/1999	Bloch	327/536	H02M 3/073
6574457	12/2002	Salminen	N/A	N/A
2008/0024223	12/2007	Park	330/277	H03F 3/16
2018/0041167	12/2017	Belkin	N/A	H03D 7/1408
2019/0020311	12/2018	Kang	N/A	H03D 7/1466

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to frequency mixing, and more particularly to a frequency mixing circuit with a multi-transistor structure.

2. Description of the Prior Art

(2) In the field of circuit design, various circuits rely on frequency mixing circuits to adjust signal frequencies to reduce or increase a frequency of an input signal to a target frequency range for the purpose of signal processing. A frequency mixing circuit is typically implemented using a single transistor. In partial structure of an external digital pre-distortion (EDPD) circuit as shown in FIG. 1, a frequency mixing circuit 10 with a single-transistor structure is utilized to perform a frequency down-conversion on signals. Specifically, a transistor M1 receives an oscillation signal LO on a gate G and mixes a signal on a drain D with the oscillation signal LO, thereby to output a mixed signal on a source S. However, such design leads to a trade-off between noise figure (NF) and counter third-order intermodulation (CIM3). That is, if it is desired to optimize the noise performance and the CIM3 performance by enlarging a size of the single transistor M1 in the frequency mixing circuit 10 (such as, enlarging the gate length/width or increasing finger number), it causes a DC offset at an output of a transimpedance amplifier (TIA) in the EDPD circuit to exceed an allowable range associated with the manufacturing process. For example, in a specific 55 nm manufacturing process, the allowable range of the DC offset should be within ± 25 mV. Therefore, there is a need to provide an innovative frequency mixing circuit to address this

problem.

SUMMARY OF THE INVENTION

(3) In view of this, it is one object of the present invention to provide a frequency mixing circuit with a multi-transistor structure, which can improve anti-interference ability against high-order harmonics without increasing sizes of the transistors. As the frequency mixing circuit with multi-transistor structure provided by the present invention ensures some transistors not to be conductive, additional noises will not be introduced excessively, such that severe degradation in the noise figure can be prevented. Since sizes of the transistors are reasonably controlled in the present invention, the undesired DC offset problem can be addressed.

(4) According to one embodiment, a frequency mixing circuit is provided. The frequency mixing circuit comprises: a first transistor and a second transistor. The first transistor has a control terminal, a first terminal and a second terminal. The control terminal of the first transistor is configured to receive an oscillation signal, the first terminal of the first transistor is configured to output a mixed signal, and the second terminal of the first transistor is configured to receive a source signal. The second transistor has a control terminal, a first terminal and a second terminal. The control terminal of the second transistor is coupled to the second terminal of the first transistor, the first terminal of the second transistor is coupled to the first terminal of the first transistor, and the second terminal of the second transistor is coupled to the control terminal of the first transistor.

(5) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates an application of a conventional frequency mixing circuit.

(2) FIG. 2 illustrates a structure of a frequency mixing circuit with a multi-transistor structure according to one embodiment of the present invention.

(3) FIG. 3 illustrates an equivalent circuit model of the frequency mixing circuit with the multi-transistor structure when the transistor is conductive.

(4) FIG. 4 illustrates an equivalent circuit model of the frequency mixing circuit with the multi-transistor structure when the transistor is not conductive.

(5) FIG. 5 illustrates an equivalent circuit model of the frequency mixing circuit with the single-transistor structure when the transistor is conductive.

(6) FIG. 6 illustrates an equivalent circuit model of the frequency mixing circuit with the single-transistor structure when the transistor is not conductive.

DETAILED DESCRIPTION

(7) In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present embodiments. It will be apparent, however, to one having ordinary skill in the art that the specific detail need not be employed to practice the present embodiments. In other instances, well-known materials or methods have not been described in detail in order to avoid obscuring the present embodiments.

(8) Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure or characteristic described in connection with the embodiment or example is included in at least one embodiment of the present embodiments. Thus, appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures or characteristics may be combined in any suitable combinations and/or sub-combinations in one or more embodiments.

(9) FIG. 2 illustrates a structure diagram of a frequency mixing circuit with multi-transistor structure according to one embodiment of the present invention. The frequency mixing circuit **100** has an oscillation signal input terminal **101**, a source signal input terminal **102** and a mixed signal output terminal **103**. The oscillation signal input terminal **101** is configured to receive an oscillation signal LO, the source signal input terminal **102** is configured to receive a source signal RF, and the mixed signal output terminal **103** is configured to output a mixed signal IF. The frequency mixing circuit **100** is configured to mix the source signal RF and the oscillation signal LO, thereby to adjust a frequency of the source signal RF. Accordingly, the frequency of the mixed signal IF can fall within a target frequency range.

(10) The frequency mixing circuit **100** includes a first transistor **110** and a second transistor **120**. The first transistor **110** has a control terminal **112** (e.g., gate), a first terminal **114** (e.g., source) and a second terminal **116** (e.g., drain). The second transistor **120** has a control terminal **122** (e.g., gate), a first terminal **124** (e.g., source) and a second terminal **126** (e.g., drain). Moreover, the control terminal **112** of the first transistor **110** is coupled to the second terminal **126** of the second transistor **120**, the first terminal **114** of the first transistor **110** is coupled to the first terminal **124** of the second transistor **120**, and the second terminal **116** of the first transistor **110** is coupled to the control terminal **122** of the second transistor **120**. On the other hand, the control terminal **112** of the first transistor **110** and the second terminal **126** of the second transistor **120** are coupled to the oscillation signal input terminal **101** of the frequency mixing circuit **100** to receive the oscillation signal LO. The first terminal **114** of the first transistor **110** and the first terminal **124** of the second transistor **120** are coupled to the mixed signal output terminal **103** of the frequency mixing circuit **100** to output the mixed signal IF. The second terminal **116** of the first transistor **110** and the control terminal **122** of the second transistor **120** are coupled to the source signal input terminal **102** of the frequency mixer circuit **100** to receive the source signal RF.

(11) In one embodiment, the oscillation signal LO has a high level HL and a low level LL. Since the control terminal **112** of the first transistor **110** receives the oscillation signal LO, these two levels of the oscillation signal LO affect a conduction state of the first transistor **110**. FIG. 3 illustrates an equivalent circuit model of the frequency mixing circuit **100** when the oscillation signal LO is at the high level HL. FIG. 4 illustrates an equivalent circuit model of the frequency mixing circuit **100** when the oscillation signal LO is at the low level LL.

(12) Regarding the equivalent circuit model shown by FIG. 3, the high level HL of the oscillation signal LO allows the first transistor **110** to enter a conductive state. Therefore, circuit characteristic between the first terminal **114** (source) and the second terminal **116** (drain) of the first transistor **110** is equivalent to a (parallel) combination of an on-resistance $R_{sub.on1}$ and a capacitive element $C_{sub.on1}$. The capacitive element $C_{sub.on1}$ formed when the transistor **110** is conductive may be further equivalent to a (parallel) combination of: 1) a capacitive element $C_{sub.b1-ON}$ and a resistive element $R_{sub.ds1}$ that are connected in series; and 2) a capacitive element $C_{sub.g1}$. In addition, since the source signal RF and the mixed signal IF have the same DC level (typically 0.5V), a voltage V_{gs} (i.e., the gate-to-source voltage) of the second transistor **120** will be 0, such that the second transistor **120** will not be conductive and become capacitive. Moreover, circuit characteristic of the second transistor **120** between the first terminal **114** and the second terminal **116** of the first transistor **110** will be equivalent to a capacitive element $C_{sub.off2}$. A capacitance of the capacitive element $C_{sub.off2}$ formed by the second transistor **120** varies with the voltage level of the oscillation signal LO. Typically, the voltage V_{ds} (i.e., the drain-to-source voltage) of the second transistor **120** will vary between $-0.5V$ to $+0.5V$, which will affect the capacitance of the capacitive element $C_{sub.off2}$.

(13) On the other hand, regarding the equivalent circuit model shown by FIG. 4, the low level LL of the oscillation signal LO allows the first transistor **110** to enter an off state. Therefore, the circuit characteristic between the first terminal **114** (source) and the second terminal **116** (drain) of the transistor **110** is equivalent to a (parallel) combination of: a capacitive element $C_{sub.ds1}$ and a

capacitive element C.sub.off1. The capacitive element C.sub.off1 is further equivalent to a (parallel) combination of: 1) a capacitive element C.sub.b1-OFF and a resistive element R.sub.ds1 that are connected in series and; 2) a capacitive element C.sub.g1. In such situation, the second transistor **120** is still not conductive and remains capacitive. A capacitance of the capacitive element C.sub.off2 varies with the level of the oscillation signal LO. During the operation of the frequency mixing circuit **100**, the capacitive element C.sub.off2 formed by the second transistor **120** can help to eliminate energy of high-order harmonics (e.g., the second harmonic (at double frequency) and the third harmonic (at triple frequency)), thereby optimizing CIM3 performance. Please refer to FIG. 5 and FIG. 6 for further understandings.

(14) FIG. 5 and FIG. 6 respectively illustrate equivalent circuit models of the frequency mixing circuit **10** with the single-transistor structure of FIG. 1. FIG. 5 illustrates an equivalent circuit model of the frequency mixing circuit **10** when the oscillation signal LO is at the high level HL. FIG. 6 illustrates shows an equivalent circuit model of the frequency mixing circuit **10** when the oscillation signal LO is at the low level LL.

(15) Regarding the equivalent circuit model shown by FIG. 5, the high level HL of the oscillation signal LO allows the transistor M1 to enter a conductive state. Therefore, circuit characteristic between a source S and a drain D of the transistor M1 is equivalent to a (parallel) combination of an on-resistance R.sub.on and a capacitive element C.sub.on. The capacitive element C.sub.on formed when the transistor M1 is conductive is equivalent to a (parallel) combination of: 1) a capacitive element C.sub.b-ON and a resistive element R.sub.ds that are connected in series and; 2) a capacitive element C.sub.g. Regarding the equivalent circuit model in FIG. 6, the low level LL of the oscillation signal LO allows the transistor M1 to enter an off state. Thus, the circuit characteristic between the source S and the drain D of transistor M1 is equivalent to a (parallel) combination of a capacitive element C.sub.ds and a capacitive element C.sub.off. The capacitive element C.sub.off is further equivalent to a (parallel) combination of: 1) a capacitive element C.sub.b-OFF and a resistive element R.sub.ds that are connected in series and; 2) a capacitive element C.sub.g. It can be understood from FIG. 5 and FIG. 6 that the frequency mixing circuit **10** with the single-transistor structure lacks anti-interference ability against high-order harmonics due to lack of the capacitive element C.sub.off2 to eliminate energy of the high-order harmonics. Hence, the CIM3 performance of the frequency mixing circuit **10** is poor.

(16) Furthermore, regarding dimension parameters of the transistors, a size of the first transistor **110** can be selected based on actual requirements of external circuits, and a size of the second transistor **120** can be selected based on the following principles to obtain better performance. In one embodiment, the size of the second transistor **120** may be smaller than that of the first transistor **110**. In one embodiment, a gate length of the first transistor **110** may be identical to that of the second transistor **120**. In one embodiment, both of the gate lengths of the first transistor **110** and the second transistor **120** may be selected according to (e.g., equal to) a minimum gate length of a manufacturing process that first transistor **110** and the second transistor **120** are produced by. In one embodiment, a gate width of the second transistor **120** may be selected according to (e.g., equal to) a minimum gate width of a manufacturing process that the second transistor **120** is produced by. In one embodiment, the gate width of the second transistor **120** may be smaller than that of the first transistor **110**. In one embodiment, a finger number of the second transistor **120** (i.e., the second transistor **120** has a multi-finger gate) may be slightly smaller than or equal to that of the first transistor **110** (i.e., the first transistor **110** has a multi-finger gate). In one embodiment, channel types of the first transistor **110** and the second transistor **120** are identical.

(17) Since the second transistor **120** remains in the non-conductive state during the operation of the frequency mixing circuit **100**, the second transistor **120** introduces capacitance only for the frequency mixing circuit **100**, which will not lead to an increase in noise. Moreover, through the capacitance provided by the second transistor **120**, the energy of the high-order harmonics can be effectively eliminated, thereby improving the CIM3 performance without introducing additional

noise.

(18) Table 1 shows comparison between performances of the frequency mixing circuit (with the multi-transistor structure) provided by present invention and the conventional frequency mixing circuit (with the single-transistor structure) with different dimension parameter selections. Specifically, regarding the frequency mixing circuit with the multi-transistor structure listed in the first column, the gate width of the second transistor **120** is 600 nm, the gate width of the first transistor **110** is 2 μ m, the finger number of the second transistor **120** is 17, and the finger number of the first transistor **110** is 19. Regarding the frequency mixing circuit with the multi-transistor structure listed in the second column, the gate width of the second transistor **120** is 600 nm, the gate width of the first transistor **110** is 2 μ m, the finger number of the second transistor **120** is 19 and the finger number of the first transistor **110** is 19. It can be found in the Table 1 that for HBNF (which regards the noise figure) the present invention does not cause excessive degradation in the noise performance when compared to the frequency mixing circuit with the single-transistor structure. In terms of CIM2 and CIM3, there is a significant improvement. Furthermore, regarding DC offset I and DC offset Q (which regards the DC offset voltage), the present invention also controls the DC offset voltage to fall within a reasonable range.

(19) TABLE-US-00001 TABLE 1

	1. Multi-transistor structure	2. Multi-transistor structure	3. Single-transistor structure
Width (nm)	600	600	2000
Gate width (nm)	600	600	2000
Finger number	17	19	19
HBNF (dB)	35.04	35.08	34.43
CIM2 (dBc)	98.52	98.51	87.35
CIM3 (dBc)	90.45	89.97	80.05
DC offset I (mV)	9.672	9.718	10.96
DC offset Q (mV)	13.29	13.38	9.176

(20) In view of above, the frequency mixing circuit **100** proposed by the present invention can effectively suppress the high-order harmonics, thereby improving the CIM3 performance. Moreover, the present invention does not lead to excessive degradation in the noise performance due to the multi-transistor structure. The present invention also solves the problem that the DC offset voltage goes higher than expected when increasing the size of the transistor under the single-transistor structure.

(21) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A frequency mixing circuit, comprising: a first transistor, having a control terminal, a first terminal and a second terminal, wherein the control terminal of the first transistor is configured to receive an oscillation signal, the first terminal of the first transistor is configured to output a mixed signal, and the second terminal of the first transistor is configured to receive a source signal; and a second transistor, having a control terminal, a first terminal and a second terminal, wherein the control terminal of the second transistor is coupled to the second terminal of the first transistor, the first terminal of the second transistor is coupled to the first terminal of the first transistor, and the second terminal of the second transistor is coupled to the control terminal of the first transistor; wherein when the frequency mixing circuit is operated to perform frequency mixing, the second transistor remains in a non-conductive state.
2. The frequency mixing circuit of claim 1, wherein a finger number of the second transistor is smaller than or equal to a finger number of the first transistor.
3. The frequency mixing circuit of claim 1, wherein a gate width of the second transistor is smaller than a gate width of the first transistor.
4. The frequency mixing circuit of claim 1, wherein a gate width of the second transistor is a minimum gate width of a manufacturing process that the second transistor is produced by.
5. The frequency mixing circuit of claim 1, wherein a gate length of the first transistor is identical

to a gate length of the second transistor.

6. The frequency mixing circuit of claim 1, wherein gate lengths of the first transistor and the second transistor are identical to a minimum gate length of a manufacturing process that the first transistor and the second transistor are produced by.

7. The frequency mixing circuit of claim 1, wherein a size of the second transistor is smaller than a size of the first transistor.

8. The frequency mixing circuit of claim 1, wherein a channel type of the first transistor is identical to a channel type of the second transistor.

9. The frequency mixing circuit of claim 1, wherein when the frequency mixing circuit is operated to perform frequency mixing, the second transistor is capacitive only.
